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Film Thickness in Starved EHL Point Contacts

This paper presents a numerical study of the effects of inlet supply starvation on film thickness in EHL point contacts. Generally this problem is treated using the position of the inlet meniscus as the governing parameter; however, it is difficult to measure this in real applications. Thus, in this paper an alternative approach is adopted whereby the amount of oil present on the surfaces is used to define the degree of starvation. It is this property which determines both meniscus position and film thickness reduction. The effect of subsequent overrollings on film thickness decay can also be evaluated. In the simplest case a constant lubricant inlet film thickness in the Y direction is assumed and the film thickness distribution is computed as a function of the oil available. This yields an equation predicting the film thickness reduction, with respect to the fully flooded value, from the amount of lubricant initially available on the surface, as a function of the number of overrollings n. However, the constant inlet film thickness does not give a realistic description of starvation for all conditions. Some experimental studies show that the combination of side flow and replenishment action can generate large differences in local oil supply and that the side reservoirs play an important role in this replenishment mechanism. Thus the contact centre can be fully starved whilst the contact sides remain well lubricated. In these cases, a complete analysis with a realistic inlet distribution has been carried out and the numerical results agree well with experimental findings.

1 Introduction

Since the fully flooded smooth EHL contact problem was solved numerically (Dowson, 1959; Hamrock, 1976), most of the research effort has been directed toward the study of non-smooth or rough surface lubrication. This is understandable as excessive roughness or local defects can lead to a severe reduction in component reliability or life, even under fully flooded operating conditions. If the EHL film is depleted due to reduced lubricant availability then even moderate surface roughness can lead to component damage. Consequently, the effect of starvation on the film thickness to roughness ratio merits further attention. This is especially true with the widespread use of greased-for-life applications, as starvation effects play a dominating role in grease lubrication (Åström, 1993; Cann, 1992). Thus, it is important to estimate the film thickness reduction correctly; in order to adjust the predicted life, the operating conditions or to improve the surface finish. On the other hand, it would also be advantageous to know the minimum quantity of oil required to adequately lubricate a contact under given operating conditions.

It was the early work of Wedeven et al. (1971), Chiu (1974), and Pemberton and Cameron (1976) that defined the starved lubrication problem. They studied starvation visually, using optical interferometry to measure film distribution within the contact and to simultaneously observe the supply condition. One observation was that as starvation proceeded an oil-air meniscus is formed in the inlet region and this moves towards the Hertzian contact circle as the severity of the condition increases. Film thickness reduction coefficients were thus derived based upon the inlet meniscus position. As a result, the supply condition was defined using the position of the inlet meniscus rather than the amount of oil present on the surfaces. The latter would be a more direct way of describing the contact behaviour as it controls both the shape and location of the inlet meniscus and the ensuing film thickness distribution.

The analytical work of Wolveridge et al. (1971) established a relation between the meniscus position and the film thickness reduction for line contacts. In numerical studies of the point contact problem by Hamrock and Dowson (1977b), the inlet meniscus position was again adopted to describe starvation. The inlet meniscus was modeled as a straight line. Formulas relating the position of the inlet meniscus to a coefficient of film thickness reduction were established. This approach can only be applied to the "classically" starved regime where the meniscus is clearly observed outside the Hertzian radius. It breaks down for the "fully" starved condition where the meniscus position approaches the Hertzian contact circle.

The application of the experimental film depletion factors to realistic cases is difficult since it is hard to establish the position of the inlet meniscus. The limitations of the numerical work are similar. The assumption of a straight line meniscus is contradicted by experimental observations (Åström, 1993; Cann, 1992; Dalmaz, 1978; Gadallah, 1984; Wedeven, 1971). More recently, it was shown (Chevalier, 1994; 1995) that a contact

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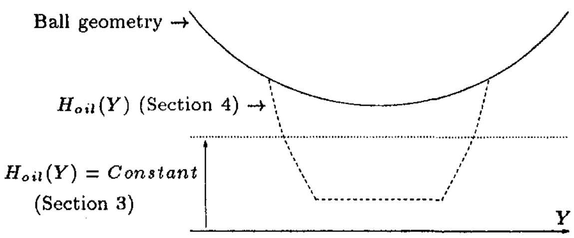

Fig. 1 Tranversal view of the inlet oil film thickness at the inlet boundary ($X = X_0$) for Section 3 and Section 4

can be locally both extremely starved and nearly fully flooded, due to large differences in *local* oil supply.

The earlier studies are therefore limited, since they cannot account for the precise geometry of the inlet meniscus. This work uses the models presented in previous papers (Chevalier, 1994; 1995) in which a physical boundary condition was used; the oil inlet film thickness distribution on the surface. It studies the influence of a constant oil film distribution in Y (see Fig. 1), and presents a general equation relating starved central film thickness and available oil film. The effect of repeated overrollings and the amount of oil displaced sideways is also studied as a function of the amount initially available. Finally, numerical and experimental results are compared for the more general case where the side reservoirs play a dominant role (see Fig. 1).

2 Theory

The theoretical formulation is based on the algorithm established by Elrod (1974; 1981). A variable θ, which represents the ratio of the oil film thickness and the gap height between the surfaces, has been included in the Reynolds equation, see Bayada et al. (1990). This variable allows conservation of couette flow in the cavitated region. Contrary to the algorithm used by Elrod, θ and P are treated as two separate variables. θ is not a representation of the compressibility effects and can not exceed 1 in the zone of full film. The problem is thus treated as a complementarity problem where ($0 \le \theta < 1$, $P = 0$) or ($\theta = 1$, $P > 0$). Consequently, the calculational domain $\mathcal{D}$ is divided in two distinct zones: pressurized and nonpressurized. The boundary between these zones, i.e. film formation and subsequent rupture, is determined automatically by the calculation. Its position depends directly on the amount of lubricant initially present on the surfaces.

2.1 Mathematical Model. The modified dimensionless Reynolds equation, which is valid in the entire domain $\mathcal{D}$, can be written as:

$$\frac{\partial}{\partial X}\left(\frac{\bar{\rho} H^3}{\bar{\eta}}\frac{\partial P}{\partial X}\right) + \frac{\partial}{\partial Y}\left(\frac{\bar{\rho} H^3}{\bar{\eta}}\frac{\partial P}{\partial Y}\right) - \lambda \frac{\partial(\bar{\rho}\theta H)}{\partial X} = 0$$

$$\mathcal{D}: X_0 \le X \le X_e,\ -Y_0 \le Y \le Y_0 \qquad (1)$$

The gap height between the surfaces is given by:

$$H(X, Y) = H_0 + \frac{X^2}{2} + \frac{Y^2}{2} + \frac{2}{\pi^2}\iint_{\mathcal{D}} \frac{P(X', Y')dX'dY'}{\sqrt{((X - X')^2 + (Y - Y')^2)}} \qquad (2)$$

Note that the lubricant film thickness is represented in each region by the θH value.

The rigid body displacement H_0 is coupled to the force balance equation which, for a circular contact reads:

$$\iint_{\mathcal{D}} P(X, Y)dXdY = \frac{2\pi}{3} \qquad (3)$$

Nomenclature

a = radius of Hertzian contact, $a = \sqrt[3]{(3wR_x)/(2E')}$
$\mathcal{D}$ = domain $X_0 \le X \le X_e$, $-Y_0 \le Y \le Y_0$
E' = reduced modulus of elasticity, $2/E' = (1 - \nu_1^2)/E_1 + (1 - \nu_2^2)/E_2$
G = material parameter, $G = \alpha E'$
h = film thickness
H = dimensionless film thickness, $H = hR_x/a^2$
H_c = dimensionless central film thickness
H_{cff} = dimensionless fully flooded central film thickness
H_m = dimensionless minimum film thickness
H_{mff} = dimensionless fully flooded minimum film thickness
H_0 = dimensionless rigid body displacement
H_{oil} = dimensionless inlet oil film thickness
H_e = mean dimensionless oil film across the Hertzian width at the outlet of the contact, $H_e = 0.5 \int_{-1}^{+1} \theta(X_e, Y)H(X_e, Y)dY$
H_{eff} = H_e fully flooded
H_i = mean dimensionless oil film across the Hertzian width at the inlet of the contact, $H_i = 0.5 \int_{-1}^{+1} \theta(X_0, Y)H(X_0, Y)dY$
$\delta_{(-1,+1)}$ = oil thickness reduction between outlet and inlet, $\delta_{(-1,+1)} = H_e/H_i$
ΔH = film thickness difference between two consecutive black lines in the pseudo interference plot
L = dimensionless material parameter (Moes), $L = G(2U)^{1/4}$
M = dimensionless load parameter (Moes), $M = W(2U)^{-3/4}$
p = pressure
p_h = maximum Hertzian pressure, $p_h = (3w)/(2a^2)$
p_0 = constant (Roelands), $p_0 = 1.98\ 10^8$
P = dimensionless pressure, $P = p/p_h$
R_x = reduced radius of curvature in x direction, $1/R_x = 1/R_{x1} + 1/R_{x2}$
R_y = reduced radius of curvature in y direction, $R_y = R_x$
r, r' = relative oil film thickness, $r = H_{oil}/H_{cff}/\bar{\rho}(p_h)$, $r' = H_{oil}/H_{cff}$
$\mathcal{R}$ = central film thickness reduction, $\mathcal{R} = H_c/H_{cff}$
u_m = mean velocity in x, $u_m = (u_1 + u_2)/2$
U = dimensionless speed parameter, $U = (\eta_0 u_m)/(E'R_x)$
W = dimensionless load parameter, $W = \omega/(E'R_x^2)$
X, Y = dimensionless coordinates, $X = x/a$, $Y = y/a$
X_0, X_e, Y_0 = dimensionless domain boundaries
z = pressure viscosity index (Roelands), $z = 0.68$
α = pressure viscosity coefficient, $[Pa^{-1}]$, $\alpha = 1.7\ 10^{-8}$
ω = external load
η_0 = viscosity at ambient pressure, $[Pas]$, $\eta_0 = 0.0089$
$\bar{\eta}$ = dimensionless viscosity, $\bar{\eta} = \eta/\eta_0$
$\bar{\rho}$ = dimensionless density, $\bar{\rho} = \rho/\rho_0$
θ = relative oil quantity, ratio of the oil film thickness and the gap height
λ = dimensionless parameter, $\lambda = (12\eta_0 R_x^2 u_m)/(p_h a^3)$

The viscosity pressure relation proposed by Roelands is used:

$$\bar{\eta}(P) = \exp\left(\frac{\alpha p_0}{z}\left(-1 + \left(1 + \frac{Pp_h}{p_0}\right)^z\right)\right) \qquad (4)$$

with

$$\frac{\alpha p_0}{z} = (\ln(\eta_0) + 9.67) \qquad (5)$$

The compressibility is taken into account with the density pressure relation proposed by Dowson and Higginson:

$$\bar{\rho}(P) = \frac{0.59\ 10^9 + 1.34Pp_h}{0.59\ 10^9 + Pp_h} \qquad (6)$$

This formulation enables us to describe any lubricant inlet profile with the following boundary conditions:

$$P(X_0, Y) = P(X_e, Y) = P(X, -Y_0) = P(X, Y_0) = 0$$

$$\text{and } \theta(X_0, Y) = H_{oil}(Y)/H(X_0, Y).$$

2.2 Numerical Techniques. The starved EHL model, including the additional parameter θ, thus becomes a free boundary problem. This additional variable complicates the numerical solution process, which is already difficult to solve. Moreover, as starvation increases the meniscus approaches the Hertzian region and the solution becomes very sensitive to its position. A relatively fine mesh is thus required to accurately predict the film thickness. Furthermore, in order to limit the influence of the boundary conditions in the y direction, it is necessary to use a large calculational domain. For these reasons a multigrid solution method has been applied to the problem and multilevel multi-integration has been used to limit the computing time of the elastic deformations. The numerical procedure is not presented in detail in this paper because it is not significantly different of those described in the following references (Brandt, 1990; Lubrecht, 1987; Venner, 1991). The complete numerical procedure is presented in detail in Chevalier (1996).

The numerical error in the central film thickness is evaluated by comparing the solution on a certain grid with the solution on a finer grid with half the original mesh size. The first-order discretization reduces the error by 50 percent between the two solutions. For the results presented, the maximum error of the central film thickness is estimated to be less than three percent.

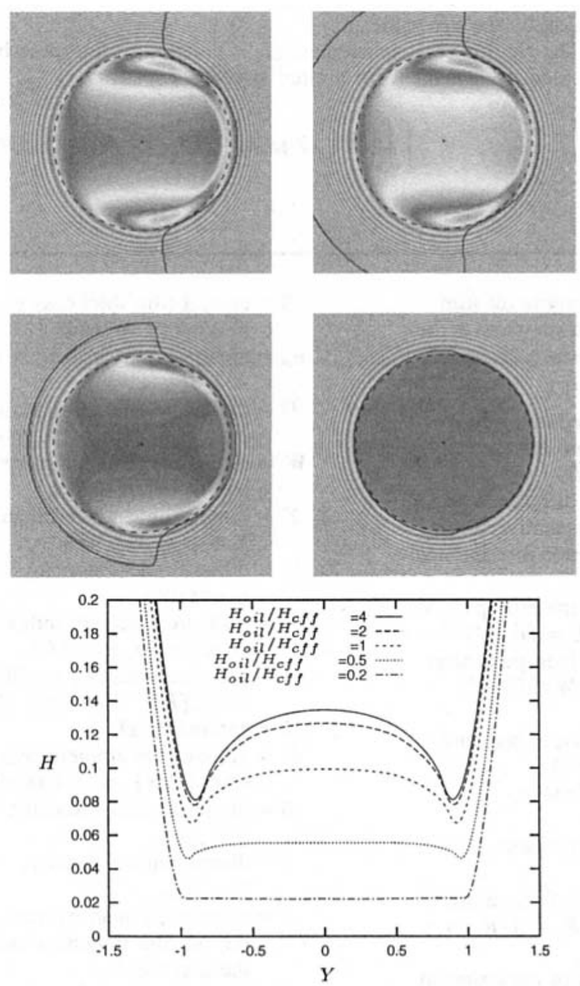

Fig. 2 M = 100 L = 10: **Pseudo-interference plots ΔH = 0.04, with film formation and cavitation boundaries, and central film thickness distribution (X = 0) as a function of H_{oil}/H_{cff}**

3 Constant Oil Inlet Film

In this section, starvation will be studied for various contact operating conditions. The oil film available between the surfaces is assumed to be constant, i.e., it does not vary with Y, $H_{oil}(Y)$ = constant. Calculations were performed on a 257*257 grid, where the domain size was adjusted according to the operating conditions: for lightly loaded cases $X_0 = -4.5$, $X_e = 1.5$, $Y_0 = 3$ and for heavily loaded cases $X_0 = -2.5$, $X_e = 1.5$, $Y_0 = 2$. In the next section, the effect of nonuniform inlet film shape is considered. Figure 1 represents the profile of the inlet oil film thickness considered in Sections 3 and 4.

3.1 Film Reduction. Figure 2 shows the pseudo-interference plots of the film thickness distribution and the central cross sections for different values of inlet oil supply. The Hertzian circle is plotted with a dashed line and the boundary of the pressurized zone with a solid line.

On the pseudo-interference plots the inlet meniscus can be observed to move toward the Hertzian circle when the inlet oil film diminishes. This meniscus conserves a simple, circular, shape concentric with the Hertzian zone. At the outlet, the film rupture location is hardly affected by the supply condition.

On the cross section plot, the central film thickness H_c is observed to decrease faster than the minimum film thickness H_m with the inlet oil film thickness H_{oil}. When H_{oil} becomes very small compared to the fully flooded central film thickness H_{cff}, the film thickness distribution tends to a flat, Hertzian shape. The ratio H_c/H_m tends asymptotically to one for all operating conditions, as starvation increases (H_{oil} decreases), see Chevalier et al. (1995) and Table 2 in Appendix I. However, under fully flooded EHL conditions the ratio H_c/H_m tends to increase with load (Venner, 1991). This means that while for increasing load the film thickness distribution tends to the Hertzian distribution in absolute terms, this is not true in relative terms.

In the study of the central film thickness H_c as a function of the amount of oil available on the surface H_{oil}, it is beneficial to divide both H_c and H_{oil} by the fully flooded central film thickness H_{cff}, as suggested by Wolveridge et al. (1971). This results in two asymptotes: $H_c/H_{cff} = H_{oil}/H_{cff}$ and $H_c/H_{cff} = 1$, since the starved central film thickness cannot exceed the thickness of the available oil layer, and secondly it cannot exceed the fully flooded central film thickness. Figure 3 shows that the relative starvation curves for very different operating conditions ranging from M = 10, L = 2 to M = 1000, L = 10 are almost identical. This is hardly surprising when one realizes that the film thickness profile of the fully flooded contact is accurately represented by the Hertzian distribution and this approximation improves as the severity of the starvation condition increases.

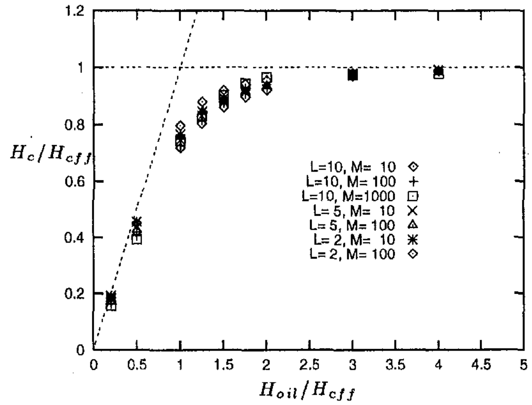

Fig. 3 Central oil film thickness as a function of the thickness of the oil inlet layer, assuming a constant inlet thickness with respect to Y

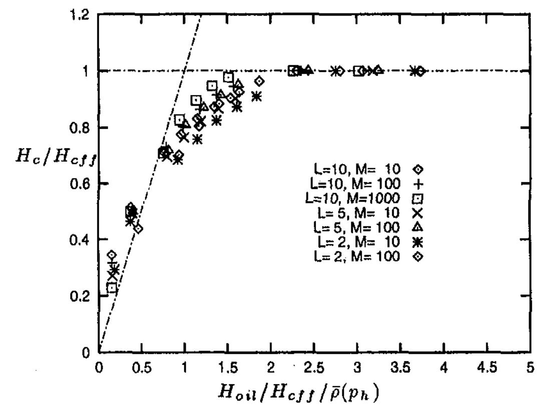

Fig. 5 Central oil film thickness as a function of the thickness of the oil inlet layer: results of Hamrock and Dowson (1977b)

However, it can be noted that for small values of the ratio H_{oil}/H_{cff}, differences between lightly and heavily loaded contacts are observed. These are due to the compressibility of the oil (see Appendix I, Table 1 and Table 2): H_{oil}/H_c has the same value as $\bar{\rho}(p_h)$ for each case, indicating that the volume flow is not conserved but the mass flow is. Consequently, dividing H_{oil}/H_{cff} by $\bar{\rho}(p_h)$, the two asymptotes are reached for any operating conditions (Fig. 4), only small variations are observed in the intermediate region.

A very simplistic analysis (see Appendix II) inspired by the work of Kingsbury (1973) leads to a general Eq. (7) relating the film thickness reduction $\mathcal{R} = H_c/H_{cff}$ to the ratio $r = H_{oil}/H_{cff}/\bar{\rho}(p_h)$.

$$\mathcal{R} = \frac{r}{\sqrt[\gamma]{1 + r^\gamma}} \tag{7}$$

Please note that $r/\sqrt[\infty]{1 + r^\infty} = \min(r, 1)$.

In the EHL regime, γ is observed to vary between 2 and 5. The fully flooded central film thickness H_{cff} can for instance be calculated according to Nijenbanning et al. (1994) or Hamrock and Dowson (1977a).

Obviously, when M increases, the behavior will follow closely the two asymptotes $\bar{\rho}(p_h)H_c = H_{oil}$ for $H_{oil} \leq \bar{\rho}(p_h)H_{cff}$ and $H_c = H_{cff}$ for $H_{oil} \geq \bar{\rho}(p_h)H_{cff}$. At the opposite end of the spectrum, for small values of M, the contact approaches the rigid isoviscous asymptote which is numerically observed to correspond to $\gamma \cong 0.5$ (see Appendix II). It is difficult to predict the behaviour in the transition region.

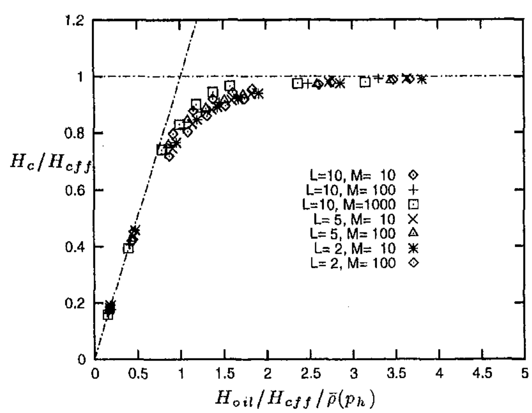

Fig. 4 Central oil film thickness as a function of the thickness of the oil inlet layer, assuming a constant inlet thickness with respect to Y. Compressibility effects are suppressed by dividing H_{oil}/H_{cff} by $\bar{\rho}(p_h)$.

3.2 Comparison With the Formula of Hamrock and Dowson. In this section, the formula developed by Hamrock and Dowson (1977b) is used to obtain the results presented in Fig. 5. To do this, it is necessary to numerically determine the position of the inlet meniscus on the central line from the assumed film thickness distribution. Comparison between Fig. 4 and Fig. 5 shows similar behaviour for $r \geq 1$. However, for small values of r, the film thickness predicted by Hamrock and Dowson exceeds the asymptotic value $\mathcal{R} = r$. For heavily starved cases, the inlet meniscus position varies very little with the supply conditions. It is thus very difficult to accurately evaluate this position. In the authors' opinion, it is easier to relate the film thickness in the contact to the inlet oil layer and they suggest that the formula developed by Hamrock and Dowson (1977b) should be used with care when the meniscus position is very close to the Hertzian zone.

3.3 Film Generation. From Fig. 4 it can be concluded that as the amount of oil available in the inlet is reduced, the ratio of $\bar{\rho}(p_h)H_c$ to H_{oil} tends to one. When studying the numerical results one finds that the meniscus position approaches the Hertzian circle, and as a consequence, pressure build-up in the inlet is delayed. Since the gradients of the Hertzian pressure distribution are very large, large pressures are rapidly attained. These large pressures induce very high viscosities, and thereby suppress pressure induced side flow. As a consequence the contact displaces less of the available oil and thus becomes more efficient in developing an EHL film.

In order to investigate this side flow behavior in detail, the average lubricant film thickness between $-1 \leq Y \leq 1$ is studied. H_i represents the mean inlet oil film thickness ($H_i = H_{oil}(Y)$ when $H_{oil}(Y)$ = constant) and H_e the mean outlet oil film thickness across the Hertzian width. The ratio H_e/H_i, i.e. $\delta_{(-1,+1)}$, is plotted in Fig. 6 as a function of H_{oil}/H_{eff}, where H_{eff} is the mean outlet oil film thickness under fully flooded conditions. From this figure it can be concluded that as the amount of oil available is less than the fully flooded central film thickness, the contact displaces less than 30 percent of the incoming oil. The less oil available, the less oil (relatively) is displaced. Consequently, the relative film thickness reduction of a ball repeatedly overrolling the same track, without reflow, slows down with every passage (see the next section).

The behavior is almost identical for all the operating conditions and it can be noted that for $H_{oil}/H_{eff} = 0.2$ only 2 to 3 percent of the incoming oil is expulsed out of the Hertzian track. This can explain why starved ball bearings at high speed still operate reliably: what is important is not the oil

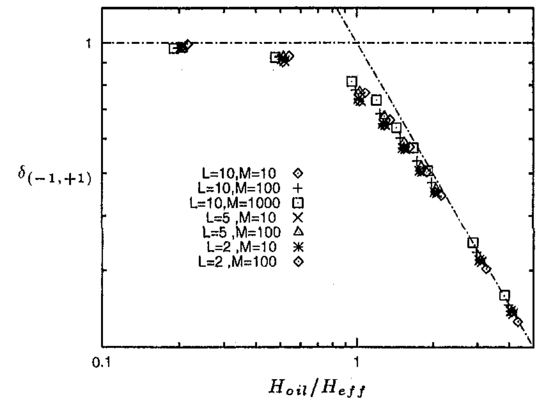

Fig. 6 Mean oil film thickness reduction in the central part of the contact due to side flow effects. $\delta_{(-1,+1)}$ represents the mean reduction coefficient between inlet and outlet.

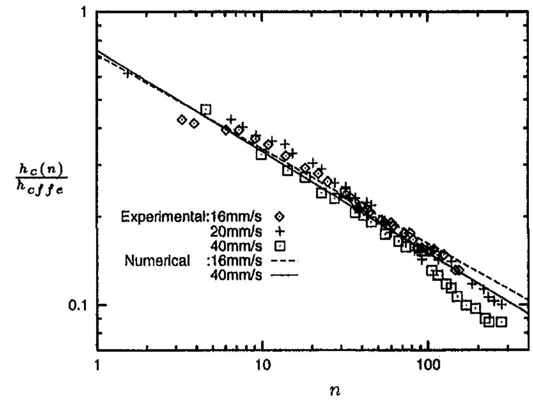

Fig. 8 Experimental and theoretical film decays. The numerical results correspond to 40 mm/s (γ = 2.9) and 16 mm/s (γ = 3.1).

quantity itself but the oil quantity compared to the fully flooded conditions. For the same oil quantity and if reflow mechanisms are neglected, the film thickness will decrease faster (other parameters remaining constant) for a contact with a speed of U than for a contact with a speed of $2U$. Figure 7 represents the evolution of the ratio $\mathcal{R}/r$ (note the similarity with Fig. 6):

$$\frac{\bar{p}(p_h)H_c}{H_{oil}} = \frac{\mathcal{R}}{r} = \frac{1}{\sqrt[\gamma]{1 + r^{\gamma}}} \tag{8}$$

3.4 Repeated Overrolling. If one considers the film thickness at the outlet to be constant over the Hertzian width, which is a realistic approximation for a starved contact, and if H_e is considered to be equal to $\bar{p}(p_h)H_c$ (see Table 1), it is possible to evaluate the film decay as a function of the number of passes n, when one supposes no replenishment action at all. Under these assumptions, $r(n) = \mathcal{R}(n - 1)$ and $\mathcal{R}(n)$ can be written as:

$$\mathcal{R}(n) = \frac{1}{\sqrt[\gamma]{1/r_0^{\gamma} + n}}, \quad \lim_{n\to\infty} \mathcal{R}(n) = n^{-1/\gamma} \tag{9}$$

Note also that

$$\lim_{r_0\to\infty} \mathcal{R}(n) = n^{-1/\gamma}, \quad \forall n$$

This result is compared to that obtained experimentally. Starved film thickness decay with repeated overrollings has been measured in a ball on disk apparatus (Guangteng, 1992).

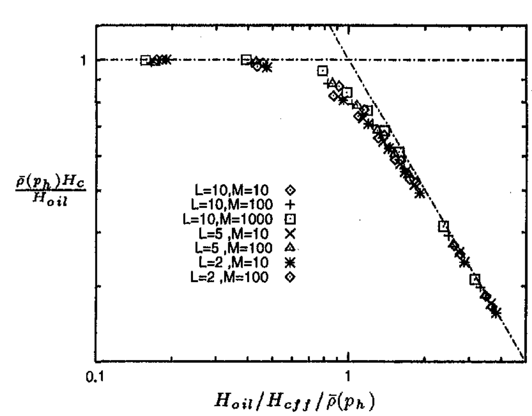

Fig. 7 Mean oil film thickness reduction: $\mathcal{R}/r$ ratio

In these experiments a thin oil film is applied to the disk and ball and the central film thickness decay with time measured for different speeds. A high viscosity ($1.37 Pa.s$) base oil was used, which reduces replenishment action between two consecutive overrollings. Numerical simulations corresponding to the speeds of 40 mm/s (M = 37.8, L = 5.15) and 16 mm/s (M = 75.2, L = 4.1) have been performed to determine the γ parameter which are, respectively, found to be γ = 2.9 and γ = 3.1. The experimental (h_{cffe}) and numerical fully flooded central film thicknesses are respectively 0.41 μm and 0.31 μm for the 40 mm/s speed and 0.24 μm and 0.17 μm for the 16 mm/s speed. The experimental and theoretical film decays, using the experimental fully flooded value as a reference, are plotted in Fig. 8. The difference between experimental and theoretical fully flooded film thickness explains why the numerical curves start at about 0.7 instead of 1.

In Fig. 8, an experimental film decay is observed with a slope of approximately −0.4, corresponding to γ = 2.5. The numerical simulations with slopes of −0.35 (γ = 2.9) and −0.32 (γ = 3.1) are represented by the lines. The theoretical predictions give a correct approximation of this phenomenon but underestimate the experimentally observed film decay which decreases more rapidly at the end of the test. It may be due to the approximated relation between H_c and H_e, which overestimates the mean film at the outlet of the contact. Moreover, the experimental decay may be accelerated by thermal effects, which can progressively modify the viscosity and hence the reference value H_{cff}.

4 Varying Oil Inlet Film

In the previous section, the oil inlet distribution was considered to be constant with respect to Y. For a contact overrolling a track for the first time, or when reflow is either dominant or absent, this assumption is realistic. However, when the contact overrolls the same track repeatedly and when reflow from the secondary reservoirs (Pemberton, 1976) only refills the outer parts of the track, this assumption is no longer valid. Hence the oil inlet film distribution can no longer be taken as constant, and its precise shape will determine the EHL oil film distribution. The center of the contact is heavily starved, while the contact sides remain nearly fully flooded (Chevalier, 1994; 1995). This leads to heavily distorted film thickness profiles, with an oil meniscus coinciding with the Hertzian circle in the centre of the contact, and then moving rapidly outwards, see Fig. 9.

In order to predict these film thickness distributions correctly, a numerical study is required which uses the inlet oil distribution. Unfortunately, it is not yet possible to measure these oil inlet profiles with sufficient accuracy. Consequently, the profile

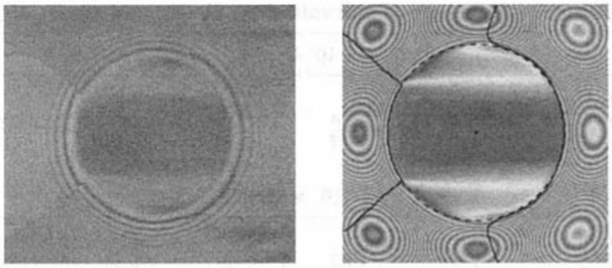

Fig. 9 Film thickness interference plot of a fully starved contact, $\Delta H = 0.02$: $M = 291$, $L = 3.28$, $\alpha = 1.5\ 10^{-8}$, $z = 0.435$, $\eta_0 = 0.0587$

was approximated by the following distribution, see Chevalier et al. (1996):

$$h_{oil} = \max\left((Y^4 - 0.1)/2,\ 0.035\right)\ \mu\text{m}$$

This approximation leads to a correct prediction of the oil inlet meniscus position and profile, and of the film thickness distribution, as can be observed from Fig. 10.

It is interesting to compare Fig. 10 with Fig. 2, starting from the central line. Around $Y = 0$ the film thickness profile is flat, and resembles that of $H_{oil}/H_{cff} = 0.5$. Around $Y = 0.4$ the film thickness starts to increase and locally H_{oil}/H_{cff} increases. Finally, at $Y = 1.0$ the film thickness shape in the side lobe region is nearly fully flooded. Therefore it can be concluded from Fig. 10 that the film thickness distribution evolves from a nearly starved condition in the centre of the contact, to a nearly fully flooded condition in the side lobe region. The contact is thus locally heavily starved and locally nearly fully flooded. In the authors' opinion, the results presented in Section 3 might be locally applied in the central part of the contact.

5 Discussion and Conclusion

This paper presents a new approach to the study of starved fluid film lubrication. In previous work the position of the inlet meniscus was used to determine the degree of EHL film reduction. This approach is limited as it cannot be used to predict film reduction for the "fully" starved condition; which is perhaps the most important regime. Also, the film thickness becomes increasingly sensitive to the meniscus position when it approaches the Hertzian circle. This distance is difficult to measure accurately for a simple test machine and almost impossible in real applications.

In the current work film thickness reduction is related to the amount of oil present on the surfaces. It is this parameter which controls lubricant flow into the contact and it also would be an easier quantity to measure in practice. The inlet film distribution has been modeled in two ways: as a constant thickness and with a variable profile.

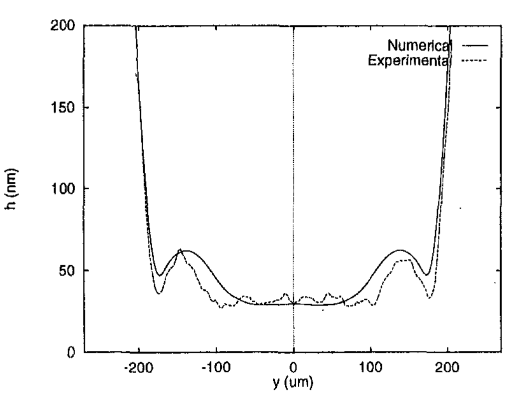

Fig. 10 Comparison of experimental and numerical film thickness profile for $X = 0$: $M = 291$, $L = 3.28$

When the film thickness is assumed constant with respect to Y, the film thickness reduction $\mathcal{R}$ from the fully flooded case depends very little on the operating conditions M, L in the EHL regime. Consequently, it is possible to devise an equation relating $\mathcal{R}$ and r:

$$\mathcal{R} = \frac{r}{\sqrt[\gamma]{1 + r^{\gamma}}}$$

For most of the EHL regime, γ lies between 2 and 5. The relative film reduction between outlet and inlet can be approximated by:

$$\frac{\mathcal{R}}{r} = \frac{1}{\sqrt[\gamma]{1 + r^{\gamma}}}$$

If replenishment action is neglected, the film decay for n consecutive overrollings can be approximated by:

$$\mathcal{R}(n) = \frac{1}{\sqrt[\gamma]{1/r^{\gamma} + n}}, \quad \lim_{n\to\infty} \mathcal{R}(n) = n^{-1/\gamma}$$

When the oil inlet thickness can no longer be considered as constant, the contact can be locally very starved and nearly fully flooded at the same time. For such conditions only a full numerical study using the precise inlet profile can completely predict the film thickness distribution in the contact.

Whether or not such a contact can operate successfully will depend on the oil inlet film level and its distribution. The oil inlet distribution depends on the frequency of overrolling and the different reflow mechanisms. The effects of lubricant reflow out of contact have not, as yet, been incorporated into the model.

The authors consider that this study is a promising approach to the problem of starved lubrication and thus, in future, it might be possible to extend it to predict starved film thickness in real life applications.

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A APPENDIX I

Table 1 presents the calculated fully flooded parameters and Table 2 the influence of starvation on H_c/H_m and H_{oil}/H_c. Please note the correlation between the $\bar{p}(p_h)$ column in Table 1 and the H_{oil}/H_c values for the last column in Table 2.

B APPENDIX II

B.1 Analytical Approach.

The analytical analysis developed here has been inspired by Kingsbury (1973). An EHL contact with an almost Hertzian deformation is considered: the side flow in the contact is supposed to be $Q = \alpha h^{(\gamma+1)}$, γ depending on the operating conditions L and M, and the volume variation (the compressibility is neglected) $dV = \beta dh$.

$$dV = -Qdt \Rightarrow \frac{dh}{h^{(\gamma+1)}} = -\alpha' dt$$

$$\frac{1}{\gamma h^{\gamma}} = \alpha' t + c$$

where t is the real time. If we consider T to be the period between two passages of the ball and n to be the number of passages, $t = nT$:

Table 1 Fully flooded parameters

	H_{cff}	H_{mff}	H_{eff}	$\bar{p}(p_h)$
$L = 10$ $M = 10$	0.687	0.510	0.671	1.15
$L = 10$ $M = 100$	0.137	0.0729	0.138	1.21
$L = 10$ $M = 1000$	0.0255	0.00883	0.0266	1.27
$L = 5$ $M = 10$	0.490	0.362	0.472	1.10
$L = 5$ $M = 100$	0.0973	0.0508	0.0946	1.16
$L = 2$ $M = 10$	0.354	0.261	0.349	1.05
$L = 2$ $M = 100$	0.0678	0.0377	0.0627	1.09

Table 2 Influence of starvation on H_c/H_m and on H_{oil}/H_c

			$L = 10, M = 10$		
H_{oil}/H_{cff}	4	2	1	0.5	0.2
H_c/H_m	1.35	1.34	1.29	1.17	1.04
H_{oil}/H_c	4.05	2.17	1.39	1.19	1.15
			$L = 10, M = 100$		
H_{oil}/H_{cff}	4	2	1	0.5	0.2
H_c/H_m	1.86	1.82	1.66	1.35	1.08
H_{oil}/H_c	4.06	2.16	1.37	1.23	1.23
			$L = 10, M = 1000$		
H_{oil}/H_{cff}	4	2	1	0.5	0.2
H_c/H_m	2.85	2.80	2.36	1.69	1.27
H_{oil}/H_c	4.08	2.07	1.35	1.27	1.27
			$L = 5, M = 10$		
H_{oil}/H_{cff}	4	2	1	0.5	0.2
H_c/H_m	1.35	1.35	1.31	1.19	1.05
H_{oil}/H_c	4.03	2.13	1.34	1.14	1.10
			$L = 5, M = 100$		
H_{oil}/H_{cff}	4	2	1	0.5	0.2
H_c/H_m	1.90	1.86	1.72	1.37	1.07
H_{oil}/H_c	4.07	2.14	1.32	1.18	1.18
			$L = 2, M = 10$		
H_{oil}/H_{cff}	4	2	1	0.5	0.2
H_c/H_m	1.35	1.33	1.29	1.18	1.07
H_{oil}/H_c	4.05	2.13	1.30	1.10	1.05
			$L = 2, M = 100$		
H_{oil}/H_{cff}	4	2	1	0.5	0.2
H_c/H_m	1.79	1.78	1.65	1.33	1.04
H_{oil}/H_c	4.04	2.09	1.25	1.11	1.08

$$n = 0,\ h = h_{oil} \Rightarrow c = \frac{1}{\gamma h_{oil}^{\gamma}}$$

If we consider the thickness after the first passage:

$$\frac{1}{\gamma h^{\gamma}} = \alpha' T + \frac{1}{\gamma h_{oil}^{\gamma}}$$

$$\lim_{h_{oil} \to +\infty} h = h_{cff} \Rightarrow \alpha' T = \frac{1}{\gamma h_{cff}^{\gamma}}$$

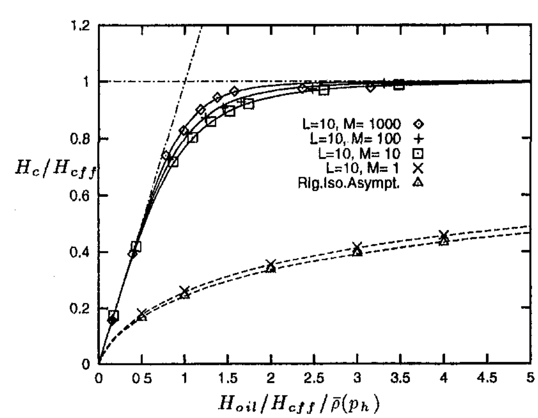

Fig. 11 Relation between $\mathcal{R}$ and r as a function of M, for $L = 10$. Evolution from $M = 1000$ to $M = 1$, tending to the rigid isoviscous asymptote: $\gamma_{1000} = 4.0$, $\gamma_{100} = 3.1$, $\gamma_{10} = 2.6$, $\gamma_1 = 0.51$, $\gamma_{Rig.Iso.Asympt.} = 0.49$

With $\mathcal{R} = H_c/H_{eff}$ and $r' = H_{oil}/H_{eff}$, it leads to a reduction coefficient:

$$\frac{\mathcal{R}}{r'} = \frac{1}{\sqrt[\gamma]{1 + r'^{\gamma}}}$$

and a central film thickness reduction:

$$\mathcal{R} = \frac{r'}{\sqrt[\gamma]{1 + r'^{\gamma}}}$$

If the compressibility is considered, r' can be replaced by r.

B.2 Numerical Results. In Fig. 11, the numerical results are plotted for $L = 10$ from $M = 1000$ down to $M = 1$. The results (with points) are compared with equation (7) in the EHL regime for $M = 1000$, 100, 10. For $M = 1$, the curve tends to the rigid isoviscous asymptote which is numerically found to be represented by $\gamma \cong 0.5$.